

BHARAT PETROLEUM CORPORATION LIMITED
Mathura Refinery — Crude Distillation Unit (CDU-1)
MAINTENANCE WORK ORDER

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Work Order No: WO-CDU1-2024-0847
Date Raised: 14-March-2024
Priority: HIGH (Urgent)
Plant Area: CDU-1 / Crude Charge Section

1. EQUIPMENT DETAILS

Equipment Tag: P-101A
Equipment Type: Centrifugal Pump (Crude Charge Pump)
Manufacturer: Kirloskar Brothers Limited
Model: KDS-200x150
Installed: 2011
Last Overhaul: 09-August-2023

2. PROBLEM REPORTED

Field operator (Shift A) reported abnormal vibration and oil leakage from the mechanical seal area of P-101A during routine 06:00 hrs round on 14-Mar-2024. Suction pressure was fluctuating between 3.8 - 4.1 kg/cm2g (normal: 4.5 kg/cm2g). Discharge temperature recorded at 142 degC (design: 135 degC). Vibration on the non-drive end (NDE) bearing measured 7.2 mm/s (alarm: 4.5 mm/s; trip: 7.1 mm/s).

3. ROOT CAUSE (PRELIMINARY)

Inspection by maintenance team indicates worn-out mechanical seal (John Crane Type 2100) and probable bearing wear on NDE. Fluid contamination suspected due to degraded seal integrity.

4. ACTION PLAN

- a) Isolate and tag-out P-101A as per SOP-CDU-PUMP-ISOLATION-Rev3.
- b) Switch over to standby pump P-101B and monitor.
- c) Replace mechanical seal and NDE bearing (SKF 6311 C3).
- d) Check shaft alignment per API 686 tolerances.
- e) Conduct trial run and record vibration signature.
- f) Update equipment history in CMMS after completion.

5. PERSONNEL

Raised by: Rakesh Sharma (Shift Incharge, Shift A)
Assigned to: Suresh Patel (Maintenance Engineer — Rotary)
Approved by: Anil Kumar (Unit Manager, CDU-1)
Safety clearance: Praveen Meena (CIP/Safety Officer, permit no. PT-2024-0319)

6. REGULATORY / STANDARD REFERENCES

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- OISD-154: Safety in Operation of Pumps and Compressors
 - API 610: Centrifugal Pumps for Petroleum Industry
 - Factory Act 1948, Schedule (hazardous processes)
 - MSDS for crude oil (document ref: MSDS-CRUDE-2023)

7. INCIDENT REFERENCE

Linked to incident IR-2024-0156 (near-miss: seal failure with no injury).

8. SPARES REQUIRED

- Mechanical seal kit, John Crane 2100 — 1 no.
- Bearing, SKF 6311 C3 — 2 nos.
- Gasket set, pump casing — 1 set

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Target completion: 16-March-2024 (within 48 hrs)

Estimated downtime cost: Rs. 4,20,000

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